Where the error of a cumulant chain lives in a deep random ReLU network

ARC White-Box Estimation Challenge 2026, Phase 2 · Algorithmic contribution write-up · team foyshal_sk · submission **#332420** (adjusted 4.090e-9, raw final-layer MSE 2.164e-8, utilisation 0.189, 0 failed MLPs) · DRAFT of 26 Sep 2026 · deadline 24 Oct 2026 23:59 UTC

Master full text: writeup/WRITEUP_DRAFT_v3.md (parity proof, measurement sections A-E, engineering, knob map); this file is the condensed deliverable. Draft status: the structure, the claims and the numbers are in place; every number is from a script in the repository (listed in section 8). Still to do: figures (error map by layer, memory horizon, cost ledger), a human editing pass, and hand-checking each derivation. Target length ≤ 10 pages.

0. Summary

We start from the public factorised third-order cumulant chain of team 504aldo (MIT), which propagates the mean, the covariance and the third cumulant (as a sum over "sources" born at each ReLU) through a 1024-wide, 16-layer bias-free ReLU network. We add one exact identity (a spherical-input reparameterisation, section 2) and cost engineering that removes 23% of the metered FLOPs for +2.8% raw MSE, mostly by exploiting the network's inactive ("dead") neurons (section 3.1): graded score 5.40e-9 (public chain) → 4.08e-9; a per-layer readout of the fourth-cumulant diagonal's pattern then takes it to **3.86e-9** (section 4.4). Most of this report is a **measured map of where the remaining error comes from, and why it resists every cheap fix we and others tried**:

1. The closures are essentially exact given true inputs. The error is made by the inputs: a perfect third-cumulant input (slice + diagonal) removes 39% of the final MSE (x0.61 on one network, x0.71 on a second), a perfect fourth-cumulant diagonal 30%, and forcing the true mean at every layer 81%. The gain is spread over layers 8-16.
2. That input error is **carried memory, not fresh physics**. Erasing all non-Gaussian history h layers back recovers only 2-6% of the deep fourth cumulant at h = 2 and 20-35% at h = 6; the third cumulant behaves the same way (keeping the sources of the last 10 layers still leaves 16% of the (2,1) slice at layer 15).
3. The carried content is **odd in the later weights**, so any statistic that is even in them (per-neuron variances, kurtosis, ensemble averages) has zero conditional correlation with it (parity theorem, section 5).
4. It is **not compressible** in the ways that would make it cheap: the elementwise products the chain needs are full-rank with a flat spectrum; no linear state closed under the dense layer maps is smaller than the symmetric cube; every contraction order of the young-source tensor network costs the same 193 square products.

1. Setting and baseline

- Task: estimate the final-layer post-ReLU means of a He-initialised bias-free ReLU MLP (n = 1024, L = 16, x ~ N(0, I)); score = MSE x max(0.1, FLOPs / 2^{41}).
- Baseline: 504aldo's V29 (github.com/504aldo/whest-p2-cumulant-k3, MIT): a port of the factorised K = 3 algorithm of Wu et al. (arXiv:2605.05179) with a memoryless, regenerated fourth cumulant, a bivariate Wick/Edgeworth closure, dense legs for sources of age ≤ 4 and a shared rank-384 (nested rank-224) basis for older sources. Public score 5.40e-9.
- Our changes (graded): Strassen block-size and first-call trims (util 0.254 → 0.244; 5.418e-9 → 5.195e-9); the spherical input reparameterisation (section 2; raw −1.1%, 5.137e-9); the feedback constant 1.25 and adaptive-lambda exponent 1.2 chosen on 100 local networks (5.128e-9); then the structural cost trims of section 3.1 (5.128e-9 → 4.090e-9).

2. An exact identity: the spherical input

Write the Gaussian input as $X = (R/\sqrt{n})\,Y$ with Y uniform on the sphere of radius $\sqrt{n}$ and $R \sim \chi_n$ independent of Y. A bias-free ReLU network h is positively homogeneous of degree 1, so $E[h(X)] = a_1\,E[h(Y)]$ with $a_1 = E[R]/\sqrt{n} = 0.99975589$ at $n = 1024$. Y has covariance I, zero odd cumulants and a small negative, exactly known fourth cumulant ($c = -2/(n+2)$ per $\mathrm{Sym}(I \otimes I)$ pair), which we inject at layer 0. Propagating Y instead of X removes the radial fluctuation from every layer's input statistics.

- Result: raw final MSE −1.1% on the grader (5.195e-9 → 5.137e-9), zero extra FLOPs.
- Credit: the idea of an angular input appears in barnobarno666's public repository; the derivation of the layer-0 fourth cumulant and its injection into the K3 chain are ours.

Hand-check: the constant c and the layer-0 core $g = -6/(n+2)$ in the estimator.

3. Where the FLOPs go

part (metered ledger of the public V29 chain, per network)	**u (1 u = 2^{31} FLOPs)**
young sources (age ≤ 4): transport of the two dense legs	54.5
young sources: (2,1)-slice contractions	50.5
joining a source to the shared basis	41-46
old sources in the shared basis	41
thin feedback legs	15
covariance, closure, Strassen additions	~20
total	**~249**

- The young tier is 193 dense square products (39 propagator transports, 54 second-leg transports, 100 slice contractions). An exact contraction-order search (forward, reverse mode, mixed, interval maps, on-demand legs, batching) finds the same 193 in every order: the elementwise (Hadamard) hyperedge must be materialised before each n^3 contraction.
- Every knob inside the chain (age gates, basis ranks, range-finder passes, feedback rank) trades about 1% cost for 1% accuracy on a 100-network paired harness: the chain sits at its own score optimum. What the knobs cannot see is the structure of the network (inactive neurons) and of the meter (rectangular Strassen families, Gram products billed at half); section 3.1 removes 23% of the cost through those two.

3.1 Structural cost trims (graded)

change	local evidence (paired)	graded adjusted (raw)
join a source to the shared basis by carrying the previous basis (no range finder)	100 networks: score x0.959, 92/100 better	4.952e-9 (#332257)
one more dense young age (5); feedback rank 16 → 4	score x0.990; x0.957 / x0.956 on two teacher networks	4.807e-9 (#332324)
drop inactive neurons from the next transport (below)	MSE −1.35% / −0.40%, util −3.4%	4.629e-9 (2.146e-8, #332391)
Strassen on the shared-basis products; Grams X d X^T as aliased Y Y^T, Y = X√d	exact to rounding, util −4.3%	4.416e-9 (2.146e-8, #332401)
(2,1) slice computed on active columns only	score x0.961 / x0.966	4.251e-9 (2.154e-8, #332404)
... and on the rows of neurons with $\Phi \geq 7{\cdot}10^{-3}$	score x0.973 / x0.961	4.090e-9 (2.164e-8, #332420)
operation budget ≤ 18k metered ops per network (grader cap of 0.4 s outside metered work); join rotation taken from the QR's R factor; feedback rank 4 → 2	exact / teacher score x0.985 / x0.995	4.0995e-9 (2.197e-8, #332626)
same, shared-basis rank 352 (19.9k ops)	Kaggle 100 networks x0.993	4.0784e-9 (#332621)

> Inactive neurons. A neuron with $\Phi(\mu/\sigma) < 10^{-2}$ (5–25% of every layer from layer 5 on) passes almost nothing forward linearly, yet its own third cumulant is essential. Measured on two teacher networks (final-MSE ratio): removing it from the next transport ($w_1 := 0$) 0.990 / 0.996; zeroing its (2,1)-slice column 1.0006 / 1.009; its row 1.011 / 1.017 at $\Phi < 10^{-2}$, but free at $\Phi < 10^{-3}$ (0.999 / 0.999) and nearly free at $7{\cdot}10^{-3}$; removing it as a hub of the newborn source x6.4–6.7 (still x1.5–1.8 at $\Phi < 10^{-5}$); zeroing its own κ_3 diagonal at intermediate layers x1.6–1.8 (+0.3% / +2.9% at $\Phi < 10^{-3}$); zeroing its covariance in the transported C x1.5–1.6 (off-diagonal only x1.15–1.2). Its mean is tail-dominated and its skew keeps propagating, so only the linear pass-through and the slice entries can go.

- The admissible drops are implemented as compaction of the contraction dimensions: live widths padded to a multiple of 32 so the rectangular families keep Strassen level 5; flopscope has no scatter, so a compacted result is expanded by one gather through an inverse index; Strassen scratch lives in one flat buffer per (role, level), so variable widths add no memory (peak commit 5.5 GiB).
- Exactness: the transport compaction agrees with the same $w_1 := 0$ without compaction to 10^{-5} relative at the final layer; the basis-side Strassen to rounding. Raw MSE moved 2.105e-8 → 2.164e-8 (+2.8%) while utilisation fell 0.244 → 0.189.

4. Where the error is made

4.1 The closures are exact; the inputs are not

- Teacher-forced closure (all inputs from a $6x10^7$-sample Monte-Carlo teacher at one layer): the post-ReLU covariance error falls to the teacher's noise.
- Channel oracles (teacher value injected at every layer, final MSE ratio, MLP #0 only so far): variance x0.53, fourth-cumulant diagonal x0.70, third-cumulant diagonal x0.87, all three x0.057.
- A perfect third-cumulant input, the (2,1) slice $\kappa_3(z_i, z_i, z_j)$ plus its diagonal: x0.608 (MLP #0) / x0.715 (a second, synthetic network). The slice alone: x0.758; the slice fed only to the chain's feedback legs: x0.9999.

- Forcing the true mean at every layer: 2.24e-8 → 4.2e-9. About 80% of the final error is mean error inherited from earlier layers.

4.2 By layer

perfect κ_3 input at layers	all	≥ 8	≥ 12	≥ 14	≤ 11 only
MLP #0	0.608	0.637	0.704	0.848	0.884
synthetic network	0.715	0.754	0.863	0.903	0.837

The windows compose multiplicatively (0.884 x 0.704 ≈ 0.62). The value is spread over layers 8-16.

4.3 The chain's own channel errors

layer	variance	C off-diagonal	κ_3 diagonal	κ_4 diagonal	(2,1) slice off-diagonal
7	0.04%	0.39%	3.2%	15%	4.6%
15	0.14%	0.73%	4.6%	32%	6.6%

The κ_4 diagonal has the right level (ratio ≈ 1.0) but the wrong per-neuron pattern (correlation 0.26-0.34 with the teacher on the neurons the mean closure is sensitive to).

4.4 A first piece of the fourth-cumulant pattern (graded)

Feeding the teacher's fourth-cumulant diagonal into the mean closure, coupled from layer 1 on public MLP #0, takes the full chain from raw 2.29e-8 to 1.61e-8. The leaders' raw MSE is 1.54–1.61e-8. The gain is the per-neuron **pattern** of the diagonal, not its level: the teacher's pattern rescaled to the chain's per-layer mean gives 1.55e-8 (108% of the gain), and a 50% blend gives 74% of it.

At every layer ≥ 6 we read the pattern linearly from quantities the chain already holds:

- the squared standardised skew $(\kappa_{3,iii}/v_i^{1.5})^2$;
- the row energy of the standardised (2,1) slice;
- the correlation-row energy $\Sigma_j C_{ij}^2/(v_i v_j)$;
- the chain's own $\kappa_{4,iiii}/v_i^2$;
- a constant.

We rescale the readout to the chain's layer mean and use it in the mean closure only.

result	value
pattern correlation with the teacher (layers 6–15, decisive neurons)	0.41–0.56 (chain alone 0.24–0.41); the coefficients transfer between networks
cost	0 FLOPs, +310 metered ops
teacher pair, raw MSE	−7.2% (fit network), −2.7% (held-out network)
graded	**3.876e-9** (#334588, ≤ 18k ops) and **3.858e-9** (#334591, rank 352); −5.4% each

Squares, products and logarithms of these features add nothing out of sample. The previous layer's features carried by $(W \circ W)^T$ add nothing either. Per-layer summaries saturate at correlation ≈ 0.5, consistent with section 5: the rest of the pattern is carried memory.

5. The error is carried memory, and it is odd

5.1 Memory horizon

history kept (layers)	1	2	3	4	6	all (linear carry)
deep κ_4 level recovered, L12	2%	6%	12%	18%	35%	83%
deep κ_4 level recovered, L16	3%	2%	5%	9%	20%	86%
pattern correlation with teacher, L16	0.20	0.41	0.59	0.65	0.77	0.80

- Method: at layer L−h, replace the pre-activation by a Gaussian with the true mean and covariance, run the true network h steps, compare the κ_4 diagonal with the teacher; "all" carries the true κ_4 of layer L−1 linearly. Fresh ReLU physics from a Gaussian input gives 1-4% of the deep level.
- Third cumulant: keeping only sources of the last 2 layers leaves 84% of the (2,1) slice at layer 15, the last 10 layers still 16%. One age mode carries ~88% of the old content (it is coherent), but six modes still leave 8.5% (it is not low-rank in age); the total κ_3 is transported exactly by one linear map, so the difficulty is spatial, not temporal.
- The next layer's κ_3 diagonal is fed 90-99% by off-diagonal κ_3, and the next κ_4 diagonal ~90% by the (2,2) pair term; even the exact repeated-index slices miss 22-29% (κ_3 diagonal) and 36-40% (slice) one step ahead: the missing part is the all-distinct content.

5.2 Parity

> The contributions of old sources to the next slice are odd functions of the later layer weights. Their conditional expectation given any statistic that is even in those weights is zero. Consequences, each confirmed by measurement: per-neuron features correlate ≤ 0.04 with the residual (102,400 neurons); marginal (copula) corrections with Gaussian dependence are worse than none; ensemble / mean-field closures cannot see the content; the fitted relation $\kappa_{4,\mathrm{off}} \approx \lambda_l\, C_{\mathrm{off}}$ captures level but not pattern.

Write out the proof (two independent derivations exist in our consultation logs) and the floor $\sqrt{(6/(r+8))}$ for expectation-only replacements; hand-check.

6. What does not work, and why

family	measured	mechanism
low-rank / Tucker / TT / CP / sketches of source legs or their products	products need 700 of 1024 directions for 99%	Khatri-Rao rank r^2, flat spectrum
any linear compressed state closed under the layer maps	(argument)	Sym^3 is irreducible under GL(n)
aggregate rank-k CP of old κ_3	unfolding rank ~250 for 2%	lower bound on CP rank
short horizons, age modes, shift-stable carriers	16% left at h = 10; 8.5% at 6 modes; 2.9%	long, coherent but not low-rank memory
diagonal-only κ_3/κ_4 transport, rank-1 κ_3, low-rank covariance	next d3 83-99% wrong; rank-1 35-49%; r = 64 still 12-33%	diagonal fed by off-diagonal content
slice-state recursions (carry the slice + a response matrix)	36% even in sample	all-distinct memory not in the slice
deterministic particles (moment-matched designs)	variance 3%, slice 74% at L15	no better than random after one ReLU
young κ_4 carrier	G4 error 0.210 → 0.203 at h = 2 (out of sample)	κ_4 memory is long
refitting the online mean correction	in-sample R^2 0.02 at depth	fresh error unpredictable from local features
layer-dependent rank reallocation	no-compression chain only −3.9% for +19% cost	error is intrinsic to births, not compression
sampling / control variates at this budget	156 u for 1.4%	MC noise floor ≫ chain bias
dropping inactive neurons as hubs, from the transported covariance, or their own κ_3 diagonal	x6.4 / x1.6 / x1.8 at $\Phi < 10^{-2}$ (section 3.1)	tail-dominated means; their skew propagates
higher inactivity thresholds for the transport drop	$\Phi < 2$–$5{\cdot}10^{-2}$: MSE +1.5–10% for util −1–2%	marginal neurons still pass signal
per-depth-block multipliers of the κ_4 constant λ	best block −1.1% locally, +1.0% graded	constant already at its optimum

7. Open problem

A method one rung more accurate than this chain must represent the all-distinct third- and fourth-order content born at every ReLU and carried across ≥ 6 layers, without the elementwise products of per-source legs. Our measurements say what such a state must reproduce (section 4), how long it must remember (section 5.1), and that no statistic even in the weights can stand in for it (section 5.2). For the fourth-cumulant diagonal the target is concrete. Per-layer summaries reach pattern correlation ≈ 0.5 (section 4.4), and the oracle says ≈ 0.7 would be worth a further 20–25% of raw MSE at no change to the third-cumulant carrier.

8. Reproducibility and LLM disclosure

- Code: the submitted estimator (#332420, submissions/v51rc7) and every probe script are in our repository; each number above names its script in our findings log. Section 3.1 variants: experiments/variants/v48_rowcompact.py, v49_strassen2.py, v50_d21col.py, v51_d21rc.py; oracles v48_coldrop*.py, v49_d21dead.py, v50_cdiag.py, v51_d3mid.py; FLOP profiler experiments/probes/flop_profile.py.

- LLM use: estimator changes, probes and this draft were written with Claude (Anthropic) in sessions directed by the participant, who chose directions, ran every experiment and checked every number. Outside consultations: Fable 5, Opus 5.5 and Sonnet 5 subagents (Anthropic), GPT-5.6 Sol and Astra 6 (OpenAI) and other assistants for ideas; their proposals were tested, and most are in section 6 as negative results. Derivations marked "hand-check" are not yet independently verified.

Before submission: freeze the submission ID (currently #332420; #332425 with row threshold 10^{-2} is grading), add figures, cut to ≤ 10 pages, human edit, post publicly on the forum or email arc-whestbench@aicrowd.com.